1. Goal of the App

This app allows users to create budgets across multiple categories and leverages AI to analyze receipts and track spending against those budgets.

In this app, users can:

- Capture receipts (photos, PDFs, manual entry)
- Itemize expenses intelligently
- Categorize spending consistently
- Track budgets over time
- Analyze spending behavior across time, location, and context

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What this app is NOT (for now):

- A bank-connected finance app.
- A net-worth tracker.

2. User Roles

Super User

Has access to all of the permissions of a contributor, plus:

- Defines budgets, categories, and settings

Contributor

- Upload receipts
- Tag receipts with context
- Views analytics, trends, and reports

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3. Core Data Concepts

Receipt

A receipt represents a single transaction.

Key fields:

- Merchant name
- Date & time
- Location (city, country; address if present)
- Original Currency
- Exchange rate (if not home currency)
- Total amount
- Taxes (explicit total if present)
- Payment method (free-form label like “MASTERCARD 1234”, “Cash”, “Venmo”)
- Line items (if available)

Receipts can be:

- Fully itemized
- Partially itemized
- Total-only

Line Item (ExpenseItem)

Each receipt may contain multiple items.

Fields:

- Description
- Quantity
- Unit price
- Total price (pre-tax)
- Assigned category
- Confidence score (AI classification)
- Optional tags (future-safe, but minimal use for now)

Items are the **atomic unit of categorization**.

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4. Categories (Single Canonical List)

We use **one detailed default category list**.

No multiple lists.

No subcategories in v1.

No custom categories yet — but the system is designed to support them later.

Default Categories (Expense-Only)

- Groceries
- Restaurants & Dining
- Coffee & Snacks
- Alcohol & Bars
- Transportation
- Gas
- Travel
- Accommodation
- Entertainment & Fun
- Shopping (General)
- Clothing & Accessories
- Electronics
- Health & Medical
- Personal Care
- Pets
- Home & Household
- Utilities
- Insurance
- Subscriptions
- Education
- Gifts & Donations
- Childcare
- Miscellaneous

Categories are:

- Mutually exclusive
- Flat
- Stable over time

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5. Taxes (Explicit Design Decision)

Tax Handling Strategy:

Proportional Allocation

Taxes are treated as a **separate expense entity**, but allocated proportionally across items.

How it works:

- Receipt lists total tax at the end
- System calculates each item's share of pre-tax subtotal
- Tax is distributed proportionally to each item
- Category totals include their allocated tax share

Why this works:

- Reflects reality without guessing item-level tax rules
- Preserves accurate category totals
- Scales later to custom tax logic if needed

Internally:

- Tax is still stored as a distinct entity (ExpenseTax)
- Allocation is deterministic and reversible

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6. Returns & Refunds

Returns are modeled as **negative expenses**, never deletions.

Item Return

- One or more line items get negative counterparts
- Category totals adjust automatically
- Original receipt remains intact

Receipt-Level Refund

- Recorded as a negative receipt
- Linked to original merchant and payment method

No income modeling in v1.

Refunds only offset expenses.

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7. Budgets

Budgets are defined as:

- Monthly
- Per category

Fields:

- Category
- Budget amount
- Optional notes

Budgets do **not** care about:

- Payment method
- Merchant
- Tax mechanics
- Contributors

They care only about:

- Category totals over time

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8. Budget Pacing / Trajectory Visualization

Concept

For variable-spend categories, the app shows:

- How much has been spent so far
- How much *should* have been spent by today
- Whether the user is ahead or behind pace

Baseline pacing model:

- Linear (day-based)
- Example: Day 15 → ~50% expected spend

Category-Level Pacing Settings

Pacing is **opt-in per category**, controlled in Settings.

Each category has:

- show_pacing: true / false

Defaults:

- Enabled for variable categories (groceries, dining, shopping, travel, etc.)
- Disabled for fixed or lump-sum categories (rent, insurance, utilities, subscriptions)

If pacing is disabled:

- Budget totals still show
- No trajectory line
- No “ahead/behind” messaging

This avoids misleading analytics.

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9. Visualizations

For pacing-enabled categories:

- Line graph:
    - X-axis: days in month
    - Y-axis: cumulative spend
- Two lines:
    - Actual spend
    - Expected spend

For all categories:

- Total spent
- % of budget used
- Remaining budget

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10. Special Budgets (Travel, Events)

Special budgets are **budget groups**, not categories.

Example:

“Ireland Summer 2026”

They include:

- Date range
- Category-level estimates (ranges allowed)
- Actual spend tracking during and after the event

Special budgets:

- Reuse the same categories
- Can enable pacing by default
- Support before/after comparisons (estimate vs actual)

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11. Payment Methods (Lightweight)

Payment methods are:

- Labels only
- User-recognized, not verified
- Optional to “enroll”

Examples:

- Cash
- MASTERCARD 1234
- Venmo
- Apple Pay

Used for:

- Filtering
- Search
- Natural-language queries

Not used for:

- Balance tracking
- Financial reconciliation

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12. Location & Context Data

If present on the receipt:

- Store address
- City / country
- Currency

Stored for:

- Geographic spending analysis
- Travel analytics
- Seasonal / weather-based insights (future)

Exchange rate is stored at transaction time for:

- Accurate home-currency reporting
- Historical correctness

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13. AI Responsibilities (Strictly Scoped)

AI does:

- OCR & parsing
- Itemization
- Initial category assignment
- Confidence scoring
- Proportional tax allocation
- Merchant normalization

AI does **not**:

- Invent categories
- Override user corrections
- Make financial judgments

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14. Scalability (Explicitly Deferred)

Intentionally out of scope for now:

- Custom categories
- HSA / FSA / tax-advantaged programs
- Asset tracking
- Income modeling
- Bank account connections

The data model supports future expansion without rework.

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15. Philosophy (Why This Holds Together)

- Receipts are immutable facts
- Interpretation layers are configurable
- Visualization is opt-in where ambiguity exists
- Complexity is postponed, not ignored
- The system favors correctness over cleverness